

Zhangchi Zhou

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EDUCATION BACKGROUND

Master in Water Resources and Hydropower Engineering 2025-Now
Wuhan University, School of Water Resources and Hydropower Engineering
Supervisors: Prof. Jiyun Song Research Direction: Urban Hydro Meteorology

Bachelor in Water Resources and Hydropower Engineering 2021-2025
Wuhan University, School of Water Resources and Hydropower Engineering
Supervisors: Prof. Jiyun Song Carbon Neutral Pilot Class GPA: 3.74/4.0

RESEARCH INTERESTS

1. Urban Land–Atmosphere Interaction:

Investigating the seasonal and diurnal variations of urban energy fluxes and CO₂ fluxes via eddy covariance measurements and wind lidar.

2. AI + Urban Hydro Meteorology:

Integrating fixed flux tower and mobile flux observations with machine learning methods to achieve high spatiotemporal mapping of urban evapotranspiration.

RESEARCH ARTICLES

1. **Zhou, Z.** and Song, J.*: Seasonal and diurnal patterns of energy and CO₂ budgets in a subtropical wetland city, Wuhan, China. Submitted to Building and Environment, under review.
2. **Zhou, Z.**, Song, J.*, Zhang, Q. : High spatiotemporal mapping of urban evapotranspiration via a hybrid physics-based and machine learning framework. In preparation.
3. Zhang Q, Song J*, Zhou Y, Zhang J, **Zhou Z.** Physics-guided deep learning resolves five-metre urban thermal stress for climate-just delivery routing. Submitted to npj Urban Sustainability.

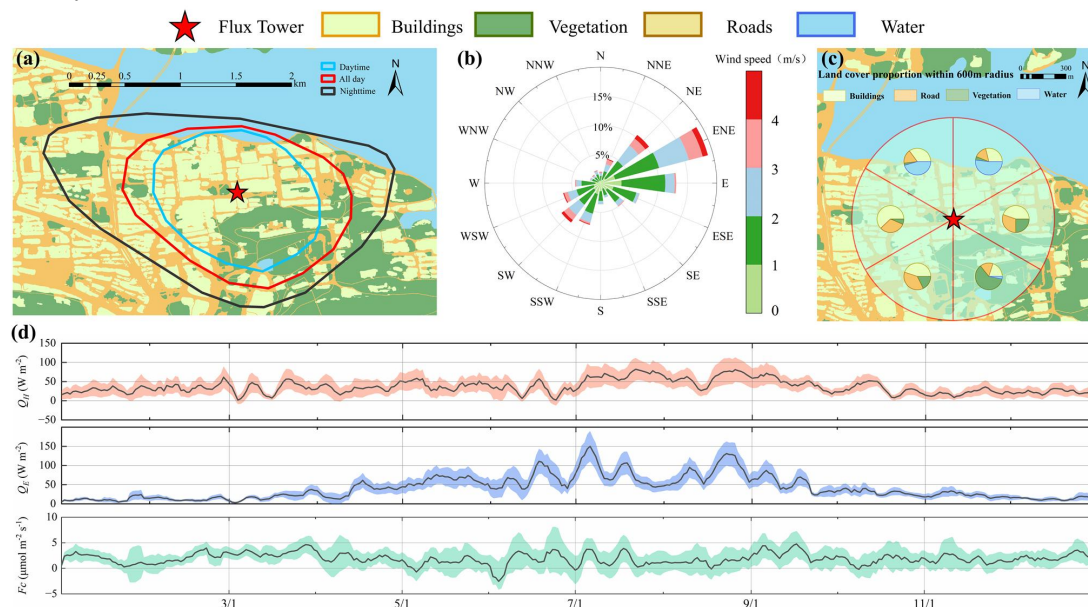
ACADEMIC CONFERENCE

1. **Zhou, Z.** and Song, J.: High Spatiotemporal Mapping of Urban Evapotranspiration via a Hybrid Physics-Based and Machine Learning Framework, EGU General Assembly 2026, Vienna, Austria, 3 – 8 May 2026, EGU26-15790, <https://doi.org/10.5194/egusphere-egu26-15790>, 2026. (Oral)
2. **Zhou, Z.** and Song, J.: Spatiotemporal Characteristics of Water, Heat, and Carbon Fluxes in the Urban Area of Wuhan, China. The 22nd China Water Forum, Xi'an, China, September 12 – 14, 2025. (Oral)
3. **Zhou, Z.** and Song, J.: Investigating urban land-atmosphere exchanges of heat, moisture, and carbon via eddy covariance measurements, 12th International Conference on Urban Climate, Rotterdam, The Netherlands, 7 – 11 Jul 2025, ICUC12-197, <https://doi.org/10.5194/icuc12-197>, 2025. (Oral)
4. **Zhou, Z.** and Song, J.: Spatiotemporal Dynamics of Urban Water, Heat, and Carbon Fluxes Via Eddy Covariance Measurements. The 10th Youth Geoscience Forum, Hefei, China, May 9 – 12, 2025. (Oral)

RESEARCH PROJECTS

1. Urban Land–Atmosphere Exchanges of Heat, Moisture, and Carbon:

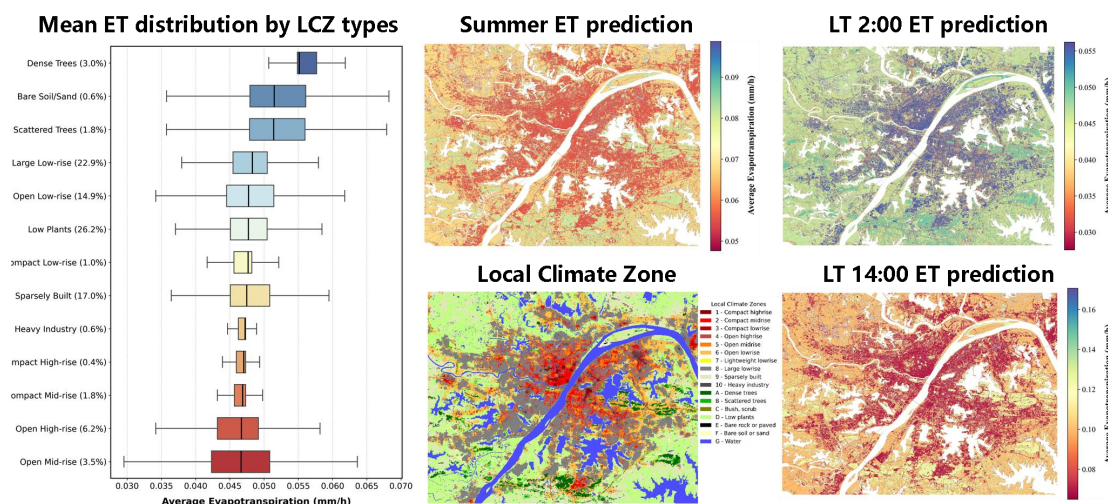
Investigating the seasonal and diurnal variations of urban energy fluxes and CO₂ fluxes via eddy covariance measurements;



Reference: **Zhou, Z.** and Song, J.*: Seasonal and diurnal patterns of energy and CO₂ budgets in a subtropical wetland city, Wuhan, China. Submitted to Building and Environment, under review.

2. High Spatiotemporal Resolution Prediction of Urban Evapotranspiration:

Integrating fixed flux tower and mobile flux observations with machine learning methods to achieve high spatiotemporal mapping of urban evapotranspiration.



Reference: **Zhou, Z.** and Song, J.: High Spatiotemporal Mapping of Urban Evapotranspiration via a Hybrid Physics-Based and Machine Learning Framework, EGU General Assembly 2026, Vienna, Austria, 3–8 May 2026, EGU26-15790,

EXCHANGE EXPERIENCE

1. Series of academic exchange activities at the University of Hong Kong (2025, completed)

We visited the Institute for Climate and Carbon Neutrality (ICCN), the Department of Civil Engineering, the Department of Mechanical Engineering, the Department of Earth and Planetary Sciences, and the Department of Geography at The University of Hong Kong (HKU), and engaged in in-depth discussions with the research groups of Professors Yuguo Li, Ji Chen, Yuyu Zhou, Jiujiu Jiao, and Peng Gong on topics related to urban hydrometeorology and carbon neutrality.

2. “Symbiosis” Global water and sustainable development Dutch and German practice research (vice captain, 2024, completed)

During this overseas practice, we visited top academic institutions such as Delft University of Technology, the Netherlands Delta Institute, and Berlin Technical University, where we engaged in in-depth academic exchanges. We also conducted on-site research at locations such as The Hague, Rotterdam New Waterway Dam, and Berlin's Potsdam Square, where we learned about the management policies and implementation of water resources and sustainable development in the local area.

HONORS AND AWARDS

1. **Outstanding undergraduate graduate** of Wuhan University, 2025
2. **Second-Class Scholarship** for Outstanding Students of Wuhan University, 2023-2024
3. Wuhan University **Shanghai Guan Long Scholarship**, 2023-2024
4. **Second-Class Scholarship** for Outstanding Students of Wuhan University, 2022-2023
5. **Second-Class Scholarship** of Wuhan University Jikang Prize 2022-2023
6. **Outstanding Communist Youth League Member** of Wuhan University, 2022、2023
7. **Outstanding Young Volunteer** of Wuhan University, 2022
8. **Third-Class Scholarship** for Outstanding Students of Wuhan University, 2021-2022
9. Wuhan University **Le Cai Scholarship**, 2021-2022

SKILLS

Eddy Covariance Data Processing: Professional software such as EddyPro and EasyFlux, covering raw data quality control, spectral corrections, and flux computation.

Flux Footprint Modeling: source area models such as Kormann & Meixner (1998) and Kljun et al. (2015).

Machine learning: LightGBM, XGBoost, Random Forest, CatBoost, SHAP.

Scientific Programming: Python, MATLAB.

Software: MS Office, ArcGIS, AutoCAD, SPSS.

Language: Chinese (native), English (CET-4: 562; CET-6: 532).